



End-to-End Data Warehouse & Analytics Project

Pakistan Stock Exchange (PSX)

Team

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Course

Data Warehousing & Business
Intelligence

Dataset Size

49,820 rows

Course Instructor

Mr. Junaid khan

Introduction: Modern Cloud Data Warehouse

Complete lifecycle demonstration using industry-leading tools.



Snowflake

Cloud Data Platform



dbt

Data Transformation &
Automation



Power BI

Business Intelligence & Analytics

Project Objectives

Key goals for transforming PSX data into actionable insights.

1 Load & Manage Data

PSX stock price data in Snowflake.

2 Build Scalable DWH

Multi-layer architecture.

3 dbt for Transformations

Automation and documentation.

4 Create Analytic Models

Fact & Dimension tables.

5 Testing & Validation

Ensure data quality.

6 Power BI Dashboards

Visual insights for decision-makers.

Dataset Overview

Daily stock information from Pakistan Stock Exchange.

Contains 49,820 rows of daily stock data (2021-2024).

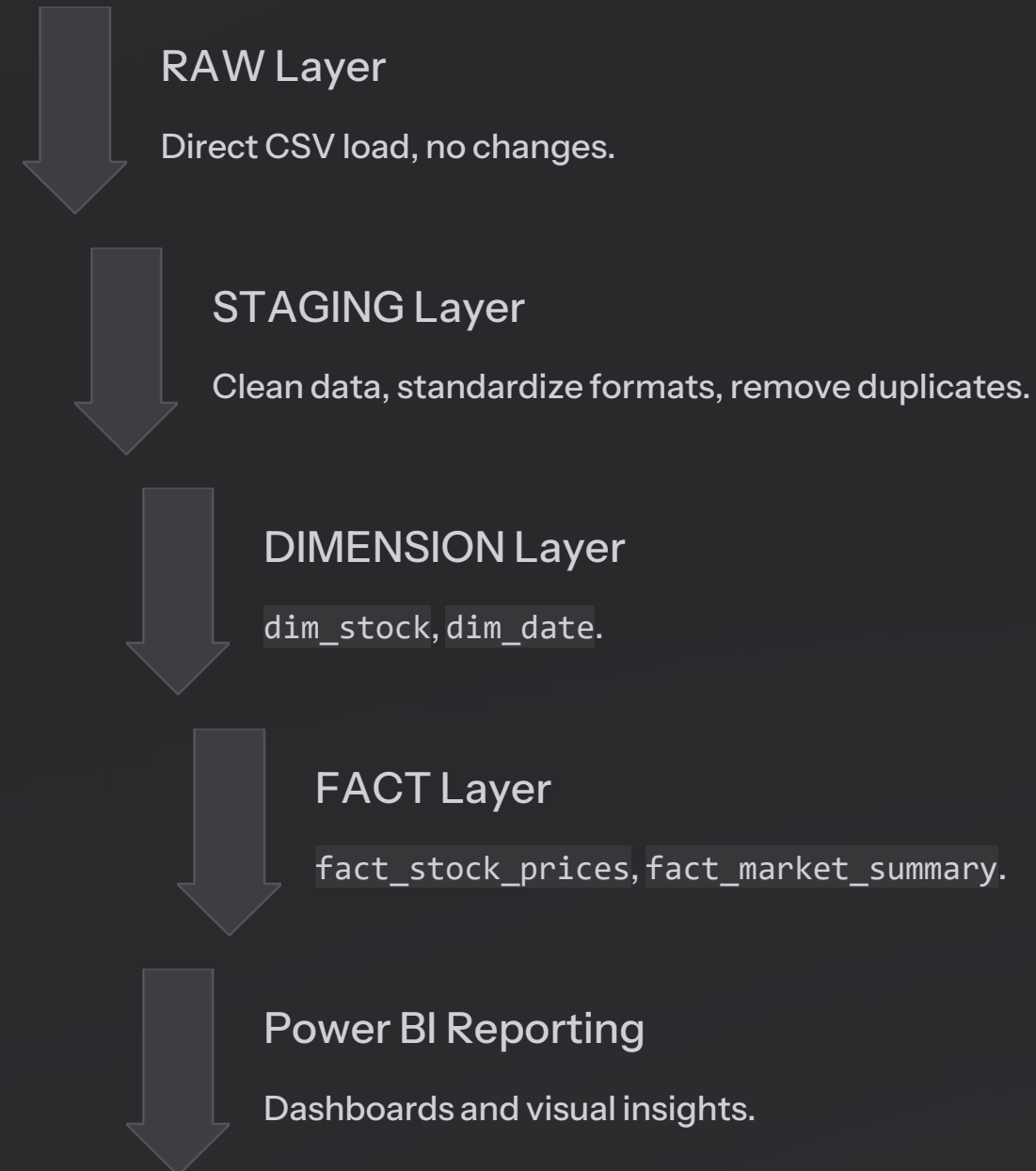
- Date
- Symbol
- Open Price
- High Price
- Low Price
- Close Price

Mostly clean, required structural standardization and date formatting.



System Architecture: Layered Approach

Industry-standard data warehousing layers.



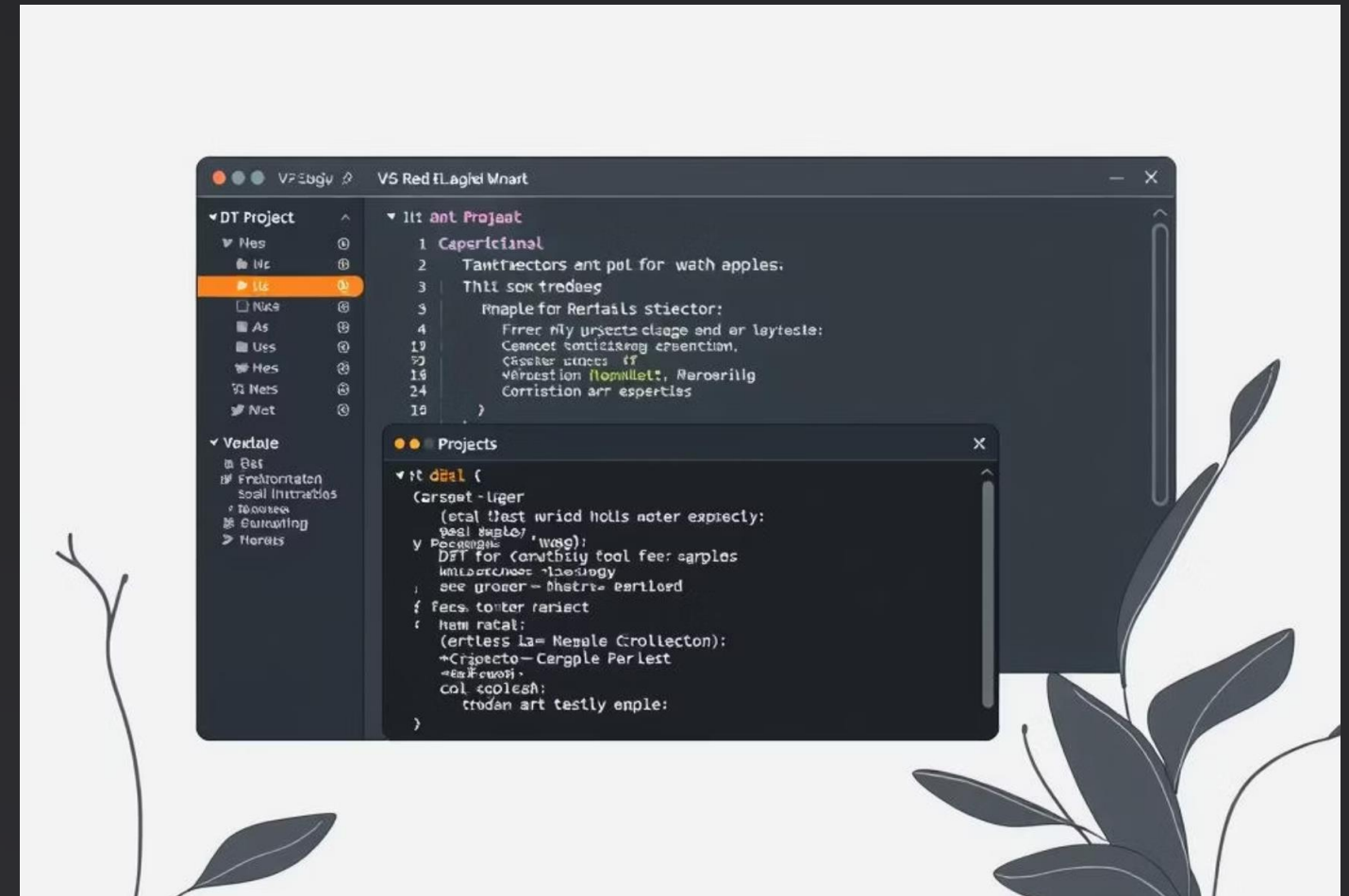
Snowflake & dbt Setup

Snowflake Configuration

- Database: PSX_DB (RAW, STAGING, DWH schemas)
- Warehouse: PROJECT_WH (Small)
- Dataset Loaded: psx_data_20211210_to_20241210.csv
- Total Rows: 49,820

dbt Project Setup (VS Code)

- Installed `dbt-core` & `dbt-snowflake`
- Configured `profiles.yml`
- Verified connection: `dbt debug`
- Folder structure: `sources/`, `staging/`, `dim/`, `facts/`, `tests/`





dbt Models: Transformation in Action

RAW Source

`psx_sources.yml` defines `PSX_DB.RAW.PSX_DATA`.



STAGING Model

`stg_psx_data.sql`: standardize, clean, convert dates.



DIMENSION Models

`dim_stock (symbol), dim_date (calendar).`



FACT Models

`fact_stock_prices, fact_market_summary.`

Data Quality & Documentation

dbt Data Quality Tests

- Uniqueness: stock_symbol, date_key
- Not Null: close_price, trade_date
- Relationship: fact-dimension integrity

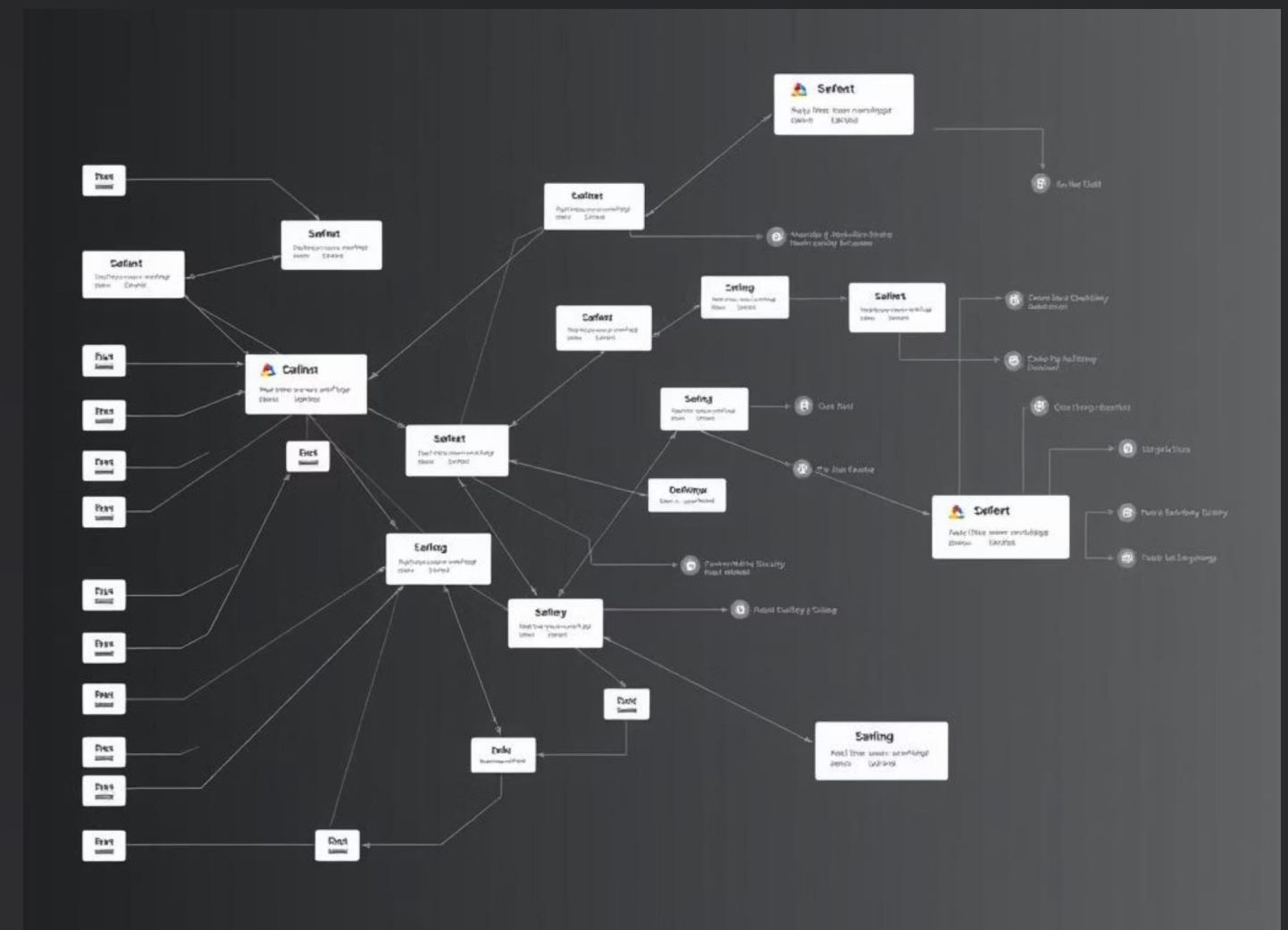
Command: `dbt test`. All tests passed successfully.

Lineage & Documentation

- `dbt docs generate`
- `dbt docs serve`

Generated: lineage graph, auto-documented models, column-level metadata, full transformation flow.

Ensures professional-grade project transparency.



Power BI: Dashboard Development

Connecting to Snowflake for visual insights.

Daily Stock Trends	Yearly Performance	Volatility Analysis	Monthly Summary
Visualize price movements.	Compare stock performance.	Identify market fluctuations.	Aggregated insights.

Key Insights & Conclusion

Key Insights Discovered

- Stocks show strong upward trends in specific months.
- Volatility increases around economic events.
- Closing prices exhibit stable patterns.
- Monthly averages aid investment predictions.



Project Success

- End-to-end cloud DWH development
- Automated ELT pipeline (dbt)
- Clean dimensional modeling
- High-quality docs & tests
- Professional Power BI dashboards

Snowflake + dbt + Power BI: Robust, industry-standard analytics stack.